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Langage

French: Native speaker (C2)
English: Professional proficiency (C1)

Formation

09/2022 - 08/2025

M.Sc. in neurosciences, neurotechnology and neurocomputation (Neuro-X)

EPFL - Neuro-X department

10/2019 - 08/2022

B.Sc. in Life Sciences Engineering

EPFL - SV department

Skills

Technical: Programming, Signal Processing, Machine Learning, Prototyping, Motion capture

Jade Therras

Engineer diploma EPF (École Polytechnique Fédérale de Lausanne)

Recently graduated, I am a polyvalent and motivated biomedical engineer with a background in **life sciences engineering** and **neurotechnology**, seeking new opportunities.
During my studies, I built expertise in programming, machine learning, neuroscience, and biomechanics, with a focus on **rehabilitation and assistive technologies.**
I'm also highly interested in **transmission and vulgarisation**, especially promoting inclusivity and equity in the scientific community.
I am looking forward to contributing to meaningful projects!

Professional & research experience

Visiting Researcher at the Motion Analysis Lab **02/2025-09/2025**
Harvard Medical School & Spaulding Rehabilitation Hospital
Boston, USA
Key words Motion capture, clinical research, MATLAB, Vicon, Visual3D, Xsens

Evaluated the use of an innovative motorised ankle-foot orthosis (Dephy Inc.) to simulate passive AFO stiffness during various ambulation tasks in post-stroke individuals. Assessed the reliability of inertial motion capture (Xsens) for estimating joint angles outside the lab, using optical motion capture (Vicon) as the gold standard.

R&D in biomechanical solutions - knee prosthetics **09/2022-02/2023**
ÖSSUR
Reykjavik, Iceland
Key words Industry, Prostetics, Solidworks, 3D printing, prototyping, python

Development of knee and foot prosthesis for amputees. I focused on prototyping a spring system to improve the stance flexion on a mechanical knee. Secondly, I developed a friction test system suitable for several knee prostheses. I prototyped the testing set-up (3D printing, aluminium), tested it and created a user-friendly computer interface. Finally, developed a screw-free modular foot pyramid.

Selection of projects

An accessible Swiss public Transport app **02/2024 - 08/2024**
EPFL, Assistive Technologies Challenge
Key words Android App, Kotlin, Assistive technology, user experience

Developed Helpie, an app to help mentally impaired people travel alone, in collaboration with Swiss public transport. This Android app guides users through their travel experience step-by-step, relying on the SBB's Swiss Mobility APIs to retrieve potential journey in real time. The app also include customable features to adapt to the customer needs and aims to gamify the travel. I was the coding leader of the app, conducting meetings with potential users and participating in project presentations. Additionally, we have been selected and have represented EPFL at an international Hackathon (EuroTeQ) in Paris.

A wearable EMG sensor for rehabilitation purposes **02/2024 - 08/2024**
EPFL
Key words clinical research, user experience, EMG, dry electrode design, Python

Aiming to design a high-quality EMG device for rehabilitation exercises, we developed the full theoretical design of a wireless, flexible microneedle-array electrode system for long-term muscle activity recording. I contributed to the global design, including electrode layout, process flow, characterisation (stress, strain, impedance), and connectors.
Post-project, we collaborated with the lab to analyse existing EMG data, performing extraction, preprocessing, and analysis in Python, and supported a funding application.

Programming Languages: *(Advanced)* Python, C++, MATLAB, R, Kotlin, LaTeX; *(Familiar)* C#, JavaScript, HTML, CSS, React ;

CAD & Simulation: SolidWorks

Motion capture: Vicon Nexus, Xsens MVM analyse, Visual3D

Software: Android Studio, Visual Studio, Unity, Godot, Excel, Overleaf, Word, Canva

Awards & Grants

2025

EPFL WISH Foundation's Fellowship

Awarded to conduct the master's thesis abroad

2024

EPFL - Certificate of recognition for teaching assistant in digital humanities

Awarded for contribution to the course of Game design and prototyping

2022

IGEM - Gold medal, Best Education Prize

As a team, with additional nominations for Best Therapeutics and Safety & Security Award

A cadmium catcher live biotherapeutic product & Game 02/2023 - 11/2023

IGEM (International Genetically Engineered Machine Competition)

Key words Game design, Synthetic biology, vulgarisation, leadership, Godot

Team project on developing a live biotherapeutic product to capture cadmium in the gut, preventing bioaccumulation. In this context, co-led the design and development of an educational video game on synthetic biology (godot). Awarded a gold medal and the best education prize. View the project website.

Also contributed to the website, preliminary research, cadmium detection (colourimetric test), safety assessment, theoretical product scaling, and regulatory aspects.

A VR Puzzle Game: "I Don't Want to Be Human Anymore" 02/2023 - 08/2023

EPFL

Key words User experience, game design, VR, Unity, C#

Developed a narrative-driven virtual reality puzzle game (Unity), featuring dual-scale gameplay (human/god modes) with real-time 3D interactions, haptic feedback, and immersive environmental manipulation to solve progressive spatial puzzles.

Teaching experience

EPFL Teaching assistant 2021 - 2024

Teaching assistants' roles included supporting students in understanding the course and practice exercises, and assisting the professor in organising the course.

- **Game Design and Prototyping** – Assisted the professor with course structure (first year of the course) and resource development, supported students through help sessions and programming guidance. The course revolve about how
- **Mathematics** – Conducted exercise sessions for four university courses: Analysis, Linear Algebra, and two first-year mathematics courses.

Animator for a mathematics course 2024

EPFL- Service de promotion des sciences

MATHeureuse allows young women to delve deep into mathematics, aiming to reduce gender inequality in scientific domains. It consists of weekend courses with numerous challenges and games. Animated courses with 2 groups of girls around 15.

Associative projects and responsibilities

- Webmaster for a theatre company - (2024-present)
- Class representative - EPFL SV Department EPFL (2021-2022)
- Co-Team leader recovery ECHO, Space Race project - EPFL rocket team (2020-2021)
- Communication manager - SV industry (2020-2021)